

Intergenerational Housing Misallocation and Fertility

Tommaso De Santo

Abstract

1 Introduction

2 Empirical Evidence

3 Model

In this Section, I develop a model of fertility and housing decisions. Agents live a finite number of periods in an overlapping generations economy. They are subject to persistent idiosyncratic income shock and transitory fertility and housing taste shock. They derive utility from consumption, housing, and fertility, and can save using a one-period liquid asset or by purchasing a home. They can either rent or own a home, and pay transaction costs when seeling the home. Owners can borrow against their home equity, subject to liquidity constraint. Households' have a fertile period of their lives during which they can attempt to have kids, which give them utility but affect their housing and consumption needs. They also have a retirement age, after which they do not work and receive pension benefits. Such benefits are financed by a payroll tax which is imposed on workers, who supply labor inelastically.

3.1 The economy

I begin by describing agents and their preferences, the income process, the markets in which agents can trade and the fertility process. Time is discrete.

Households and Preferences. The economy is populated by overlapping generation of households, who live for at most T periods. Households are also denoted as 'adults'. Adult age is denoted by $a \in [0, J]$. Agents can die at each period, and I denote s_a their probability of surviving to next period. Households work before retirement age a_R and can attempt births during the fertile window $a \in [a_F^0, a_F^1]$. Let n_t denote children ever born and m_t children currently at home. Children are dependents and do not work. Each dependent child matures and leaves the household with a fixed probability fix each period, independently of the other children. Maturation reduces m_t but does not change n_t . Households value consumption, housing services, and children at home, an discount future utility by β . At any period t , agents have flow utility

$$u_t(c_t, s_t; m_t) = \frac{\mathcal{C}(c_t, s_t; m_t)^{1-\sigma}}{1-\sigma} + \psi_t m_t.$$

The parameter $\sigma > 0$ governs curvature over the consumption aggregator $\mathcal{C}$, with the logarithmic limit understood when $\sigma = 1$, while ψ_t measures the direct utility benefit of a child at home. For a given number of children, $\mathcal{C}$ is increasing and concave in consumption and housing services. Children increase the household's needs, so $\mathcal{C}_m < 0$, and raise the marginal contribution of housing services to the aggregator, $\mathcal{C}_{sm} > 0$. The functional form of $\mathcal{C}$ is specified in the quantification.

Households either rent or own their residence. An owner occupies h_t units of housing, while a renter occupies h_t^R units. Housing services are

$$s_t = \mathbf{1}\{\text{own}\}\chi h_t + \mathbf{1}\{\text{rent}\}h_t^R, \quad \chi > 1. \quad (1)$$

The parameter χ captures the additional services households obtain from owning a home of a given size.

At death, households value the estate they leave. Let w_{t+1}^e denote bequeathable wealth, comprising liquid wealth and housing value. Bequest utility is

$$B(w_{t+1}^e) = \theta_0 \frac{(\theta_1 + w_{t+1}^e)^{1-\sigma}}{1-\sigma}. \quad (2)$$

The parameter θ_0 scales the strength of the bequest motive, while θ_1 shifts the wealth level at which it is evaluated; the logarithmic limit again applies when $\sigma = 1$.

Fertility. During fertile ages, a household chooses whether to attempt a birth. An attempt succeeds with age-specific probability π_a and, upon success, increases both n_t and m_t by one. The first birth carries a one-time utility cost ξ . Households observe extreme-value taste shocks ε_t^F before choosing whether to attempt; their scale is κ_1 for the first birth and κ_C for subsequent births. Birth success is then realized. Households subsequently observe housing taste shocks ε_t^H and choose housing and saving conditional on the realized number of children. Thus, housing can adjust to a birth within the period.

Production and income. Competitive firms produce the consumption good using effective labor L_t with constant productivity A :

$$Y_t = AL_t, \quad w = A. \quad (3)$$

The wage per unit of effective labor therefore equals labor productivity. A working household supplies $e_a z_t$ units of effective labor, where e_a is the age-earnings profile and z_t exposes the household to idiosyncratic earnings risk. Earnings are taxed at rate τ^w . Retirees receive a pay-as-you-go pension ϖ_t , financed by current payroll taxes. Disposable income is

$$y_t^d(a, z_t) = \begin{cases} (1 - \tau^w) w e_a z_t, & a < a_R, \\ \varpi_t, & a \geq a_R. \end{cases} \quad (4)$$

Households also receive a lump-sum transfer T_t from property-tax revenue, which is rebated equally across households.

Housing markets. A unit of housing trades at price P_t . Aggregate housing supply is

$$H_t^s = H_0 P_t^\eta. \quad (5)$$

Here H_0 sets the scale of supply and η is its price elasticity. Owned housing can be as large as $\bar{h}$, whereas rental housing is limited to $\bar{h}^R < \bar{h}$. The largest homes are therefore available only through ownership. Housing depreciates at rate δ_H and is subject to a property tax at rate τ_t^p . Selling an owner-occupied home costs a fraction ψ^s of its sale value.

Competitive landlords rent housing at a unit rent r_t . A landlord purchases at P_t , pays depreciation and property tax, and resells at P_{t+1} . Equating the return to the gross bond return R_b gives

$$r_t = (R_b + \delta_H + \tau_t^p) P_t - P_{t+1}. \quad (6)$$

Rent covers financing, depreciation, and property taxes, net of the change in the value of the house.

Saving and mortgage finance. Households save in a one-period bond with gross return R_b . Borrowing is secured by housing: an owner can borrow up to a fraction ϕ of the home's purchase value, while renters cannot borrow. Buying a home requires a down payment of $(1 - \phi) P_t h_t$. The

household must finance this payment out of beginning-of-period liquid wealth and the proceeds from selling its previous home, net of selling costs. Current earnings and transfers arrive after the housing transaction and cannot finance the down payment. Households that retain their home do not make a new down payment.

4 Quantification

5 Policy

6 Conclusion

References